        - Use models like Wide & Deep or embeddings from transformer architectures for scalability.

4. Performance Evaluation:
    - Split data into training and test sets for validation.
    - Use evaluation metrics such as:
        - Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) for rating predictions.
        - Precision, Recall, and F1-score for ranking accuracy.
        - Hit Rate or Coverage to evaluate the diversity of recommendations.

5. Deployment:
    - Develop an API using frameworks like Flask or FastAPI to serve recommendations.
    - Integrate the API with the e-commerce platform's front-end for real-time user interactions.
    - Deploy the model on cloud platforms like AWS or Google Cloud for scalability and performance.

6. Advanced Features:
    - Personalize recommendations based on contextual information such as time, location, and device.
    - Implement a hybrid system combining collaborative and content-based filtering for improved accuracy.
    - Add a feedback loop to learn from user interactions and improve suggestions.
    - Include explainability features to show users why certain products are recommended.

7. Optimization and Scalability: